

Name Student ID.....



Faculty of Engineering
King Mongkut's Institute of Technology Ladkrabang
Final Examination, 1st Semester, Academic Year 2020
Subject: 13016384 .Database.Systems.... Year:3.....
Examination Date: Wed December..1.,2021.....Time: ..13.30.-.16.30.....

Instructions

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1. Answer all the questions on your paper sheets. After you finish, please submit the exam answers to suphamit.ch@kmitl.ac.th.
2. Books, lecture notes, and internet access are allowed. No communications or file sharing allowed. This exam has 6 questions, 100 marks. There are 2 pages including this page.

1. Explain the meaning of the following terms: (25 Marks)

- 1.1 Wait-Die Deadlock Prevention Scheme
- 1.2 Intention Lock Modes
- 1.3 Write Timestamp (Q)
- 1.4 Write Ahead Logging
- 1.5 Checkpoint

2. Answer the following questions:

- 2.1 What is the Slowly Changing Dimension (SCD) problem? Please explain and give examples.
- 2.2 What are the solutions to the Slowly Changing Dimension problem? Please explain and give examples.
- 2.3 Why are surrogate keys used in Data Warehouses instead of the conventional primary keys from the data sources? (15 Marks)

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3. Explain Distributed Database Architecture and corresponding transparencies. Demonstrate how to write SQL codes for both Select and Update at each level of transparency. (15 Marks)

4. 4.1 Given the following Relational Schemas and the following SQL statement:

S(S#, SNAME, CITY, STATUS) SP(S#, P#, QTY) P(P#, PNAME, COLOR)

SELECT SNAME

FROM S

WHERE S.S# in (Select S#

FROM SP

WHERE P#='P5')

AND CITY='BANGKOK'

Please show 2 possible execution plans in the relational algebra. (10 Marks)

4.2 Write an SQL statement and a relational algebra statement for the query "Who supply all parts? List SNAME." (10 Marks)

5. Explain how the Multiversion 2-Phase Locking Technique works. Also explain how the technique solves the 4 concurrency control problems. (15 Marks)

6. Explain database recovery problem statements and how log-based recovery solves the problems. Also explain how log-based recovery works without the AIJ, advantages and disadvantages of this setting. (10 Marks)

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